David Pratama Widjaja, EIT

RENEWABLE ENERGY ANALYST | SOFTWARE DEVELOPER

widjaja.david@ymail.com | david-pw.com | DavidWidjaja-137 | david-pw

Skills

Power System Analysis NREL SAM, PvSyst, Signal, Fault & Event Analysis, Load Flow Calculations, pvlib, PyPSA, SLDs & Electrical Schematics

Analytics & Data Science Python (Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, etc), PySpark, Tableau, MATLAB

Software Development Python (Flask, Django, SQLAlchemy), Go, SQL, gRPC, HTML/CSS/JS, VueJS, Plotly Dash

Cloud & DevOps Amazon Web Services(AWS Certified Developer - Associate), Heroku, Terraform, Linux, Shell, Git, CircleCI

Mechatronics SolidWorks, OnShape, LTSpice, Mechanical & Electrical Assembly, 3D printing

Embedded Software C, Arduino, STM32, TI microcontroller, RF & Cellular modems, CAN bus analysis

Work Experience

Recurrent Energy (Canadian Solar O&M)
Guelph, Canada

System Performance Analyst II — *Sept 2024 - Present*

- Created NERC GADS reporting system, managed all GADS reporting across north american PV and BESS portfolio. Assembled monthly technical reporting for PV, BESS and substations.
- Migrated solar analytics tooling to AWS. Promoted python best practices and standards within organization.
- Interpreted O&M agreements and SLDs, created process for calculating, verifying, and auditing contractual availability for substations. Assisted in onboarding sites to our management.
- Refactored and standardized all contractual availability calculations to a common framework

Clir Renewables
Vancouver, Canada

Renewable Analytics Software Developer — *Apr 2023 - May 2024*

- Supported the automation of a time-consuming, weekly internal process into a workflow on AWS Glue, leveraging Terraform IaaS, CircleCI, and Amazon Athena integrations with Tableau Server. Supported reliability improvements in data quality and anomaly identifications in Python and Go.
- Implemented a tool to benchmark performance across Clir's portfolio of solar farms, enabling the company to deliver market insights to clients.
- Implemented, using Plotly Dash, an internal tool to analyze, store, and visualize signal trending anomalies and power curve changes across hundreds of wind turbines, allowing analysts to identify and surface issues for clients.

Clir Renewables
Vancouver, Canada

Renewable Energy Analyst — *June 2022 - Apr 2023*

- Significantly improved the internal tooling system for solar analytics processes, by automating repetitive tasks, improving the code efficiency of several batch jobs by an order of magnitude, and generalizing our internal PV models to adequately represent a wide variety of farm configurations. Providing these incremental improvements allowed a rapidly-growing number of urgent and specific client requests to be delivered ahead of formal product releases, while maintaining an agile system for prototyping improvements.
- Designed and implemented several detectors to automatically classify inverter and combiner-level anomalies into discrete IEC conditions, which were accepted as features into the production Clir system.
- Supported the delivery of regular and ad-hoc client requests for analysis and reporting, led several client calls on their asset performance.

Clir Renewables
Vancouver, Canada

Renewable Energy Analyst Co-op — *May 2021 - Aug 2021*

- Developed and tested several anomaly detection and classification algorithms to quantify energy losses in utility-scale solar power plants, leveraging the use of NREL's PySAM and PvSyst models. These algorithms are used to provide rapid updates to clients about the performance of their assets, resulting in improved energy yields and rate of returns.
- Designed and implemented a flexible platform with Python, SQL and AWS infrastructure to accelerate the process of developing new anomaly detection and classification algorithms.

Dometic
Richmond, Canada

Product Designer - Mechatronics — *May 2021 - Aug 2021*

- Verified the functionality of safety critical firmware for marine vehicle control systems, under a tight release schedule. Maintained and updated several test benches for this purpose.
- Assembled electromechanical test benches and waterproof telemetry systems and supported the design and verification of networked vehicle electronic control units.
- Developed and tested firmware in C for an interactive Linux-based display used to configure and monitor ECUs on a CAN network.

Schneider Electric *Burnaby, Canada*

Solar Predictive Analytics Design Technician Co-op *Jan 2019 - Apr 2019*

- Designed and implemented an interactive data synchronization/document management tool in Python, SQL and various web technologies to accelerate the process of writing and approving safety-critical maintenance guides for utility-scale solar power inverters.

Education

The University of British Columbia *Vancouver, Canada*

BaSc in Engineering Physics *2017 - 2022*

International Major Entrance Scholarship

Projects

Guelph Tool Library *Guelph, Canada*

Volunteer Librarian *January 2025 - Present*

Assisted the community in maintaining and operating the local tool library, as a means to economic empowerment and environmental sustainability.

Personal Project *Vancouver, Canada*

Indonesian Power Systems Analysis Project *May 2024 - Present*

Ongoing effort to build an open source model of the Indonesian Grid, leveraging the PyPSA toolkit. Studied utility plans, government regulations, as well as think tank and NREL publications.

Personal Project *Vancouver, Canada*

Personal Website & Blog *May 2022 - Present*

Designed my own personal website and blog using GitHub Pages and the Jekyll templating engine.

Personal Project *Vancouver, Canada*

Personal Finance Tracker *Oct 2023 - May 2024*

- Implemented, using Plotly Dash, a data analysis tool to aggregate and categorize my personal spending patterns across all my financial accounts
- Designed and implemented in Django and VueJS a public web app to upload, update, categorize, and retrieve financial transactions.

UBC IEEE Power & Energy Society *Vancouver, Canada*

Co-chair of the UBC IEEE Power & Energy Society *Sept 2018 - Aug 2019*

- Co-organized the Power & Energy Mixer 2018, a social-networking event bringing together students and industry professionals from prominent energy corporations to discuss pressing issues pertaining to the future of renewable energy in British Columbia.
- Handled logistics, marketing and digital media considerations relating to the Power & Energy Mixer 2018 throughout the event-planning stage.
- Coordinated and delegated appropriate tasks to a small team of volunteers.
- Helped to improve public knowledge of the energy sector, as well as the implications of climate change on modern society.

UBC Engineering Physics *Vancouver, Canada*

Robotic Leg Capstone Project *Aug 2020 - Apr 2022*

- Designed, using OnShape, a test stand which constrains and measures the motion of a robotic leg around spherical coordinates. Managed the bill of materials and led assembly of all mechanical components.
- Adapted an existing robot leg design for our use case, using OnShape. Led electrical and mechanical assembly of the robot leg.
- Developed an understanding of three-phase FOC control for induction motors. Researched the dynamics of underactuated, legged robots using the SLIP (Spring-Loaded Inverted Pendulum) model.

UBC Engineering Physics *Vancouver, Canada*

Engineering Physics Autonomous Robotics Competition 2 *January 2020 - May 2020*

Developed a Python program to perform localization and movement of a robot within a ROS environment, using the NumPy and OpenCV libraries.

UBC Engineering Physics *Vancouver, Canada*

Engineering Physics Autonomous Robotics Competition *May 2019 - Aug 2019*

- Designed and manufactured a chassis for a robotic drivetrain. Helped to tune the PID algorithm to enable line-following capability.
- Developed an algorithm to translate cylindrical coordinate designations into actuator positions for a robotic arm with 4 degrees of freedom.

UBC Solar Car Engineering Design Team *Vancouver, Canada*

Aeroshell Team Member *Aug 2020 - Feb 2021*

- Conducted bending analysis and material selection for carbon-fibre and core materials that will constitute the composite matrix for a new aeroshell for a new vehicle.
- Designed, using Solidworks, removable mounts for the wheel fairings of a solar-powered vehicle
- Contributed to the manufacturing of a fiberglass mold with a wet-layup procedure and a carbon fibre aeroshell with a vacuum-infusion procedure.

UBC Solar Car Engineering Design Team *Vancouver, Canada*

Software Team Lead *Sept 2019 - Aug 2020*

- I started the development of an integrated vehicle environment simulator which analyzes real time solar irradiance, geographic information system (GIS) and weather data to improve our racing strategy.
- I led the ongoing effort to complete the telemetry system and to debug the firmware on the current solar vehicle.
- I trained new members of UBC Solar in embedded software development.

UBC Solar Car Engineering Design Team *Vancouver, Canada*

Software Team Member *Sept 2017 - Aug 2019*

- Started development of a wireless telemetry system to facilitate real-time data transfer from our solar vehicle to a data visualization website, using 900 MHz radio modules and cellular chips.
- I co-wrote bare-metal firmware for the vehicle dashboard, the fault warning lights, and a speed control system on the STM32F1 micro-controller.

References available upon request.